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Diffusion-Based Trajectory Planning for Excavators with Learned Dynamics Models

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Abstract

Diffusion-based trajectory planners have shown remarkable ability to generate diverse, conditioned trajectories for robotic systems, yet they typically rely on a low-level controller to translate state-derivative commands into control inputs. For complex machines like hydraulic excavators, the low level controller design and collecting extensive hardware demonstration pose significant challenges. In this work, we collapse the plan-then-track paradigm into a single framework that generates state trajectories jointly with the control signals required to execute them. We leverage learned dynamics models to simulate large-scale demonstration datasets and to enforce dynamic coherence during generation, eliminating the need for both analytical modeling and costly real-world data collection, bridging the gap between diffusion based planning and tracking control learning.

Keywords: Trajectory generation, learned dynamics, diffusion, planning, control

1. Introduction

Excavators are complex hydraulic machines whose dynamics are highly machine-specific, nonlinear, and difficult to model analytically. Collecting demonstrations on physical hardware is intensive, since an operator must simultaneously coordinate multiple links (boom, arm, bucket) in a coupled hydraulic system. Traditional approaches separate plan generation from tracking control: a planner produces a desired state trajectory, and a separate controller maps it to actuator commands. Recent developments in trajectory generation using diffusion models are capable of generating trajectories that adhere to certain conditions and costs (Janner et al., 2022; Carvalho et al., 2023). In this work, we aim to develop a trajectory generation pipeline using demonstrations simulated with the learned dynamics of the excavator.

2. Methods and Contributions

Learned Forward and Inverse Dynamics. Inspired by prior work on dynamics modeling (Spinelli et al., 2024), we train four LSTM models over a rolling history of joint angles, joint velocities, control inputs, and pressures for the boom, arm, and bucket: (1)**Induced Pressure Predictor** ($(\text{hist}_{t-9:t-1},\, u_t) \to \hat{p}_t$), (2)**Induced Velocity Predictor**($(\text{hist}_{t-9:t-1},\, u_t,\, \hat{p}_t) \to \hat{\dot{q}}_t$), (3)**Required Pressure Predictor**($(\text{hist}_{t-9:t-1},\, \dot{q}_t^*) \to p_t^*$) and (4)**Required Control Predictor** ($(\text{hist}_{t-9:t-1},\, \dot{q}_t^*,\, p_t^*) \to u_t^*$)

Simulation and Dataset Generation: We build a Gazebo simulation integrating the learned models with the excavator URDF. Given a reference plan in joint angles and velocities, the inverse models predict the required pressure and control inputs while the forward models predict induced pressure and velocity. The full state-action trajectory (joint angles, velocities, pressures, and valve commands) is logged, enabling scalable dataset generation without real hardware.

History-Conditioned Diffusion Planner and Reward-Guided Sampling: We formulate a conditional denoising diffusion process conditioned on the history of past observations, the desired start position q_s, and the desired goal position q_g. Each trajectory waypoint comprises joint angles q, joint velocities $\dot{q}$, extend/retract pressures p for all three joints, and control inputs u. Inspired by Diffuser (Janner et al., 2022), we introduce a reward signal that biases sampling toward dynamically feasible trajectories. For diffusion step i and waypoint t, the total reward is given by:

$$\mathcal{J}^i_{\text{total}} = \sum_{t=1}^{T} -\lambda_1 \|\dot{q}^i_t - \hat{\dot{q}}_t(\text{hist}_t, u^i_t, \hat{p}_t)\|^2 - \lambda_2 \|p^i_t - \hat{p}_t(\text{hist}_t, u^i_t)\|^2$$

The gradient at timestep i and the sampling mean update is given by:

$$g^i_t = \underbrace{\nabla_{[\dot{q}^i_t, p^i_t, u^i_t]} \mathcal{J}^i_t}_{\text{waypoint } t} + \underbrace{\sum_{k=1}^{\min(10, T-t)} \nabla_{[\dot{q}^i_t, p^i_t, u^i_t]} \mathcal{J}^i_{t+k}}_{\text{next history window}} \qquad \tilde{\mu}^i_{[\dot{q}_t, p_t, u_t]} = \mu^i_{\theta[\dot{q}_t, p_t, u_t]} + \alpha \cdot \Sigma^i \cdot g^i_t$$

At every sampling step, start and goal positions are inpainted (Janner et al., 2022).

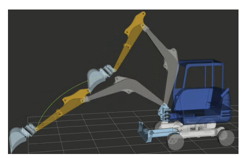
3. Preliminary Results

Dynamics Model Accuracy. Table 1 shows prediction metrics of the learned forward and inverse dynamics models. The models closely track the ground-truth joint velocities, pressures and control commands across all three joints, validating their use as a simulation backbone.

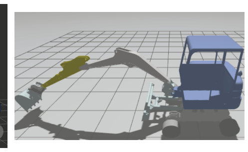
Table 1: Comparison of LSTM Metrics and Model Architecture

Pressure (Fwd)(MPa)				*Pressure (Inv)(MPa)*				*Vel(rad/s)/Cmd*			
	MAE	**RMSE**	R^2		**MAE**	**RMSE**	R^2		**MAE**	**RMSE**	R^2
arm_ext.	0.178	0.461	0.997	arm_ext.	0.293	0.717	0.992	dq_arm	0.011	0.038	0.965
arm_ret.	0.263	0.676	0.989	arm_ret.	0.317	0.763	0.985	dq_boom	0.004	0.014	0.949
boom_ext.	0.126	0.344	0.993	boom_ext.	0.259	0.631	0.980	dq_buck	0.016	0.050	0.972
boom_ret.	0.195	0.640	0.990	boom_ret.	0.321	0.739	0.989	marm	0.022	0.067	0.974
buck_ext.	0.116	0.419	0.990	buck_ext.	0.116	0.310	0.991	mboom	0.025	0.076	0.949
buck_ret.	0.189	0.445	0.995	buck_ret.	0.168	0.530	0.988	mbuck	0.012	0.067	0.924

Plan in MoveIt2



Excavator in Gazebo



Simulation Environment and Planned Trajectory. Figure 1 shows the Gazebo simulation environment and Figure 1 the excavator URDF and a trajectory planned in MoveIt.

Discussion: Moving forward, our efforts are focused on generating multi-link articulated demonstrations to build the dataset. This data will be used to train a diffusion model conditioned on historical embeddings to capture the initial dynamics of the system. The model will then be used to iteratively sample trajectories via reward-guided ODE sampling and execute the initial control until the target is reached.

References

João Carvalho, An T. Le, Mark Baierl, Dorothea Koert, and Jan Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1916–1923, 2023. doi: 10.1109/IROS55552.2023.10342382.

Michael Janner, Yilun Du, Joshua Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. 05 2022. doi: 10.48550/arXiv.2205.09991.

Filippo Spinelli, Pascal Egli, Fang Nan, Thilo Bleumer, Patrick Goegler, Stephan Brockes, Ferdinand Hofmann, and Marco Hutter. Reinforcement learning control for autonomous hydraulic material handling machines with underactuated tools. pages 12694–12701, 10 2024. doi: 10.1109/IROS58592.2024.10802199.